DENDROCHRONOLOGY

Dating the past and reconstructing environments

Dendrochronology Lesson 1
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Today’s lesson

0–10 min	Thinking about the order of events
10–25 min	Why dates matter and what dates mean
25–45 min	An overview of dating methods
45–60 min	Archives of past environments
60–80 min	Exercise in choosing samples and methods
80–90 min	Reporting ages and reviewing the lesson

Questions linking history and environment

When did it happen?

Place building repairs, forest changes and climate variation in time.

Which came first?

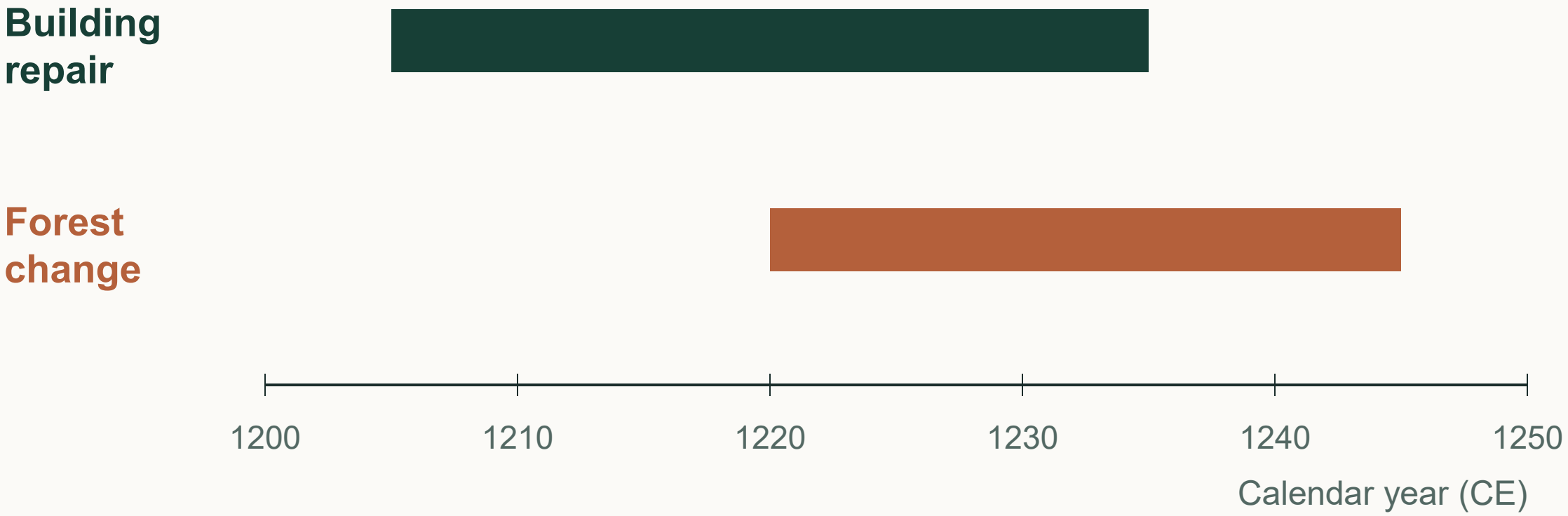
Order and time lag affect the explanations we can test.

What can the evidence show?

Distinguish simultaneous events from a causal explanation.

Date ranges and the order of events

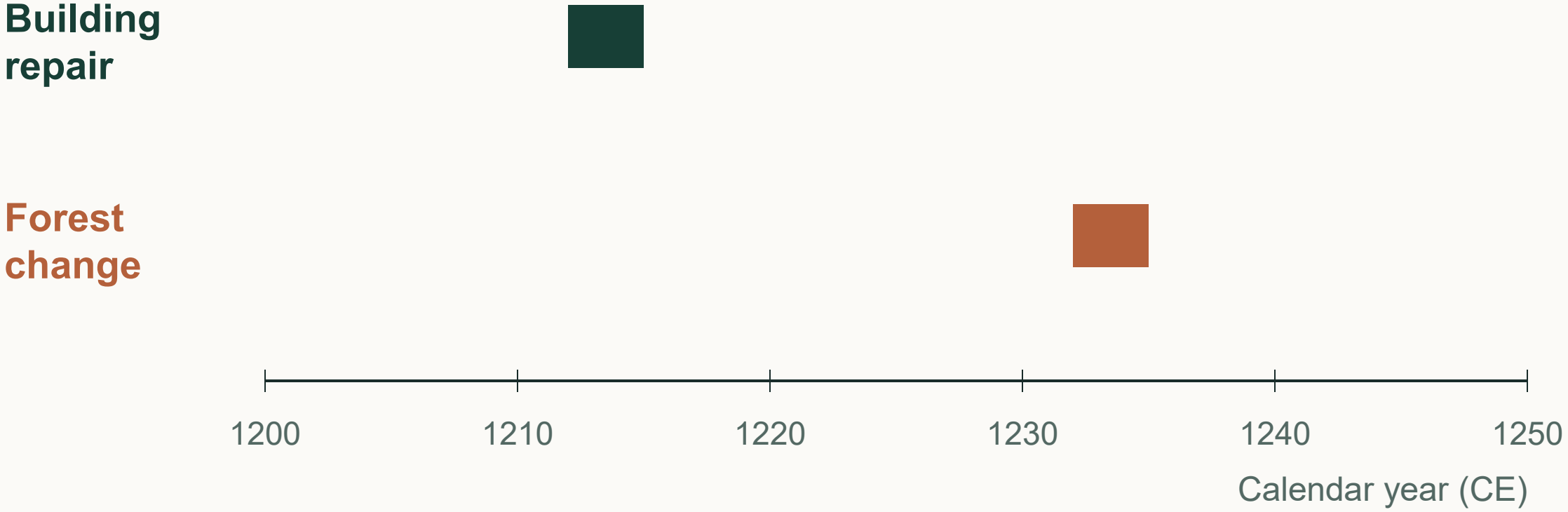
Overlapping date ranges can leave the order unresolved.



Schematic example. Ranges do not specify a confidence level.

Relations revealed by tighter dates

Tighter dates help us examine order, time lag and duration.



Schematic example. Ranges do not specify a confidence level.

Why study the past?

Understanding history

Compare documents, objects and structural timbers.

Putting the present in context

Compare current conditions with variability beyond instrumental records.

Informing decisions

Use histories of change to inform forest management and preparedness.

Relative and numerical ages

Relative age

An older-or-younger relationship

Example: lower and upper strata

Check for overturning and redeposition.

Numerical age

An age expressed in years

Example: a ring formed in 1300 CE

Numerical dates retain assumptions and uncertainty.

Sample age and event age

The year a ring formed can differ from the year a building was constructed.

Ring formation
1200 CE

→

Felling
1300 CE

→

Construction
1320 CE

→

Reuse
1450 CE

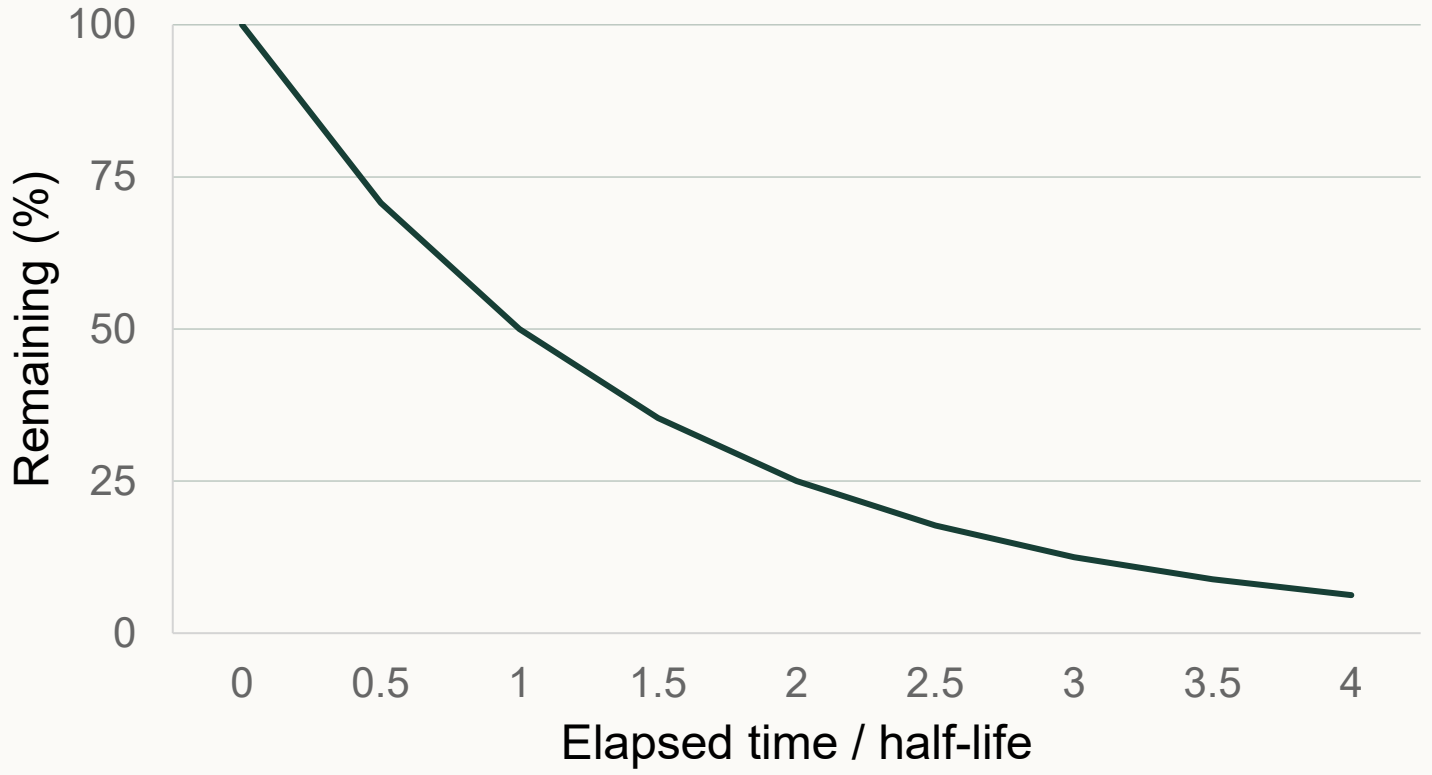
Relate the dated ring to the event of interest.

Schematic diagram. Time intervals are not to scale.

Dating methods and their targets

Method	Sample or record	Dating evidence
Dendrochronology	Wood with growth rings	Matching ring patterns
Radiocarbon	Wood, seeds, bone	Radiocarbon in the sample
K–Ar / Ar–Ar	Minerals in volcanic rocks	Decay and argon accumulation
U–Th	Speleothems, for example	Uranium–thorium isotope relationships

Radioactive decay and half-life



100%
50%
25%
12.5%

After 0, 1, 2, 3 half-lives

After each half-life, half of the parent isotope remains.

Radiocarbon dating

Sample

Examine carbon in suitable biological material, such as wood or seeds.

Measurement

Use radiocarbon decay, while considering contamination and carbon sources.

Interpretation

Calibrate the radiocarbon age and interpret a calendar-age range.

Dating rocks and carbonates

K–Ar / Ar–Ar

Suitable minerals in volcanic rocks.

Consider clock closure and retention during cooling.

Dating ash requires suitable minerals and depositional context.

U–Th

Suitable carbonates, including speleothems.

Use isotope changes after formation.

Assess initial components and later exchange.

Dating with light, heat and magnetism

Luminescence dating

Measure stored signals in minerals.

OSL uses light resetting; TL commonly uses heating.

Assess resetting and the burial dose-rate environment.

Archaeomagnetic dating

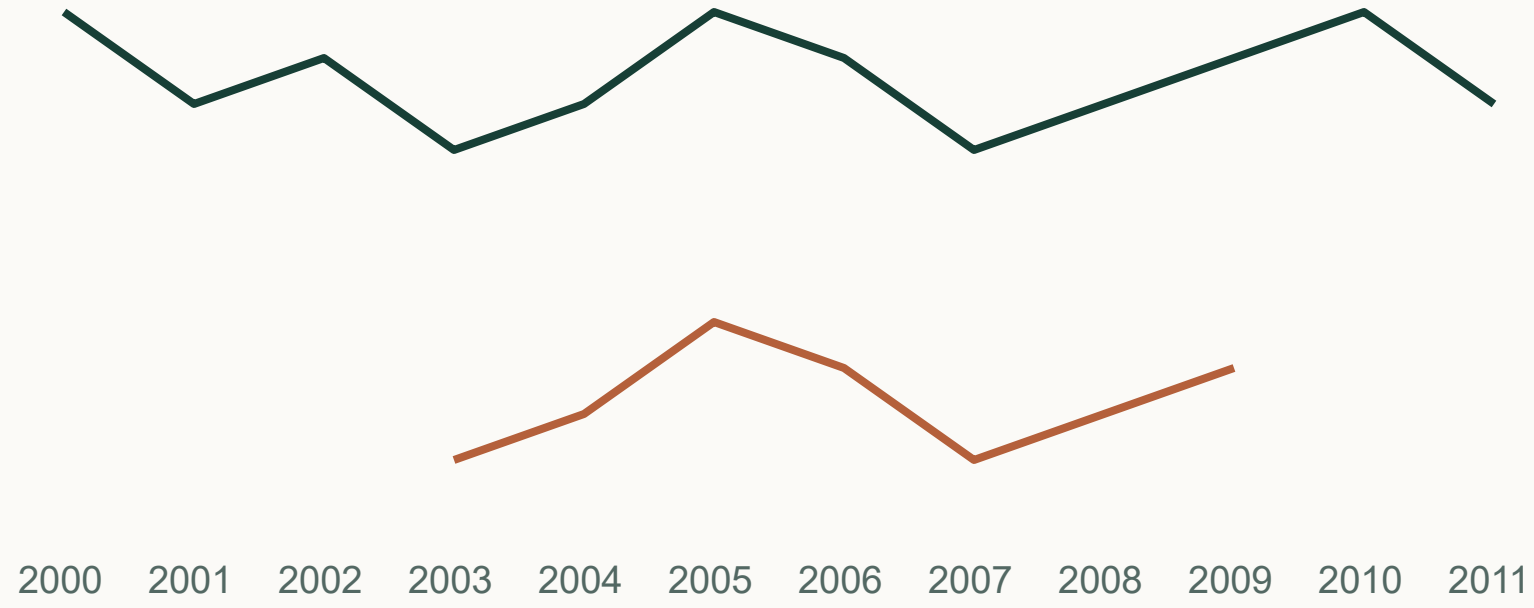
Examine remanent magnetism in fired materials.

Compare cooling-related magnetism with a regional reference.

Check movement after firing and the site location.

The role of dendrochronology

Match patterns of wide and narrow rings to assign calendar years.



Schematic data. Widths use arbitrary units.

Archives of past environments

Archive	Measurements	Environmental evidence
Tree rings	Width, density, isotopes	Conditions experienced by trees
Ice cores	Isotopes, dust, gas bubbles	Climate and atmospheric records
Lake sediments	Pollen, chemistry, varves	Vegetation and catchment history
Corals	Skeletal chemistry, growth bands	Marine environmental records

Interpreting proxy evidence

Compare with observations

Establish which conditions the proxy responds to.

Consider multiple influences

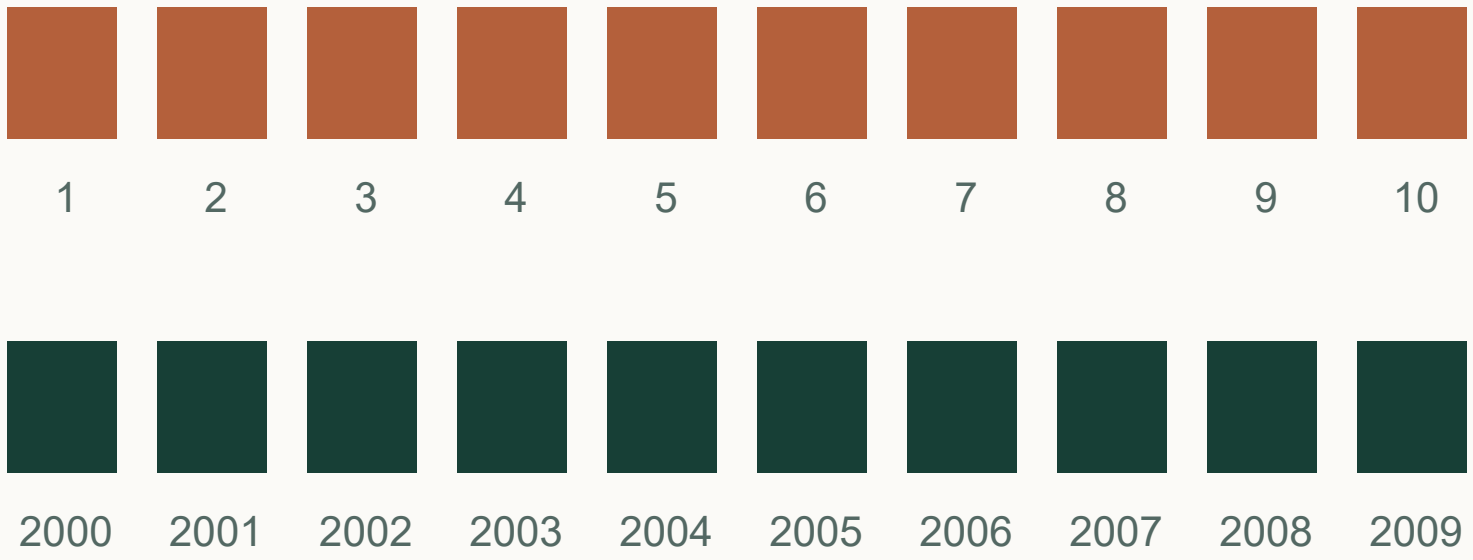
Ring width can reflect age and competition as well as climate.

Check the range of application

Relationships can vary with region, season, species and timescale.

Temporal resolution and age uncertainty

A record may resolve individual years while its calendar position remains uncertain.



Schematic diagram. Fine resolution differs from calendar anchoring.

Comparing independent records

Align the timescales

Check years, seasons, calibration and reference years.

Compare the same variable

Early-spring temperature differs from summer moisture conditions.

Investigate mismatches

Consider uncertainty, response lag, regional variation and preservation.

Exercise Choosing samples and methods

1 **Which event do you want to date?**

2 What property of the sample will you measure?

3 Which method fits, and what does its date represent?

4 What additional information is needed?

Exercise samples

ID	Available material	Target event
A	Charred seeds from a hearth	Date the activity involving fire.
B	A timber with its outer wood removed	Date construction or repair.
C	Fired material in an apparently undisturbed kiln	Date the last firing.
D	Buried wind-blown sand	Date deposition of the sand.

Reasons for choosing a method

Match the sample and clock

Which evidence fits: carbon, rings, magnetism or mineral signals?

Identify the dated event

Does the date concern formation, cooling, burial, felling or use?

Specify further checks

Consider contamination, missing outer rings, movement, resetting and context.

Interpreting the exercise

ID	Possible method	Meaning and further checks
A	Radiocarbon	Dates seed carbon. Assess its association with hearth use.
B	Dendrochronology	Dates surviving rings. Assess missing outer wood and reuse.
C	Archaeomagnetism, for example	Dates a cooling-related record. Check regional references and movement.
D	OSL, for example	Dates burial after light resetting. Assess the completeness of resetting.

BP and calendar-age notation

1000 ± 30 BP	Example conventional radiocarbon age, referenced to 1950. Do not convert it to a calendar year by subtraction alone.
cal BP / cal BCE / cal CE	Calibrated calendar ages. Record the curve and assumptions. cal BP also refers to years before 1950.
1300 CE	Example of a ring’s calendar year. The historical CE/BCE calendar has no year zero.

Key points

Choose methods for the sample and question

Different samples from one site can date different events.

Separate dating from environmental interpretation

Establish when a record formed, then investigate what it records.

State the conditions of an interpretation

Combine dates, context and independent evidence.

Finding reliable information in English

Key terms

dendrochronology

crossdating

Search phrases

dendrochronology crossdating tutorial

radiocarbon calibration sample context

Reading priorities

Definitions, captions, samples, timespans and references

Check the institution, date and links to original research.

Further reading and open data

Introduction to dating	Smithsonian Human Origins Program Introductory explanation
Radiocarbon methods and results	University of Oxford ORAU Methods and primary explanation
Scientific dating methods	Historic England Methods and primary explanation
What are proxy data	NOAA NCEI Introductory explanation
International Tree-Ring Data Bank	NOAA NCEI Open data
Glossary of Dendrochronology	WSL Terminology and methods

Next lesson Tree growth and ring structure

What does the vascular cambium produce?

How do earlywood and latewood differ?

Which features identify an annual boundary?